DATA TYPE

To store anything in our system, we should have to allocate the memory [bytes]. This memory comes with 3 properties and they are decided by the data type.

1. Type of data we want to store in that memory
2. No of bytes required
3. Value range

int sal = 32000;

unsigned sal = 65000;

long int sal = 300000;

C & C++ 16 bit compilers → 2^{16} → 65536/2 = 32768

C / C++ **int** → -32768 to 32767 ← signed int / short int ← 2 bytes ← %d ← conversion char / format specifier

Unsigned int → **0 to 65535** → 65536 → %u → 2 bytes

Signed long int → -2147483648 to +2147483647 → 4 bytes → %ld

Unsigned long int → 0 to 4294967295

Java / .net / py → int → 32 bit compilers → 2^{32} → 4294967296/2 = 2147483648

Java / .net / py → -2147483648 to +2147483647

unsigned int

long int

real numbers: decimal no's

float → 4 bytes → %f → $3.4 * 10^{-38}$ to $3.4*10^{+38}$

long float / double → 8 → %lf → $1.7*10^{-308}$ to $1.7*10^{+308}$

long double – 10 → %Lf → $3.4*10^{-4932}$ to $1.1*10^{+4932}$

char city[6] = "Hyd-16";

characters 2 types

1. Signed char → -128 to +127
2. Unsigned char → 0 to 255

   %c → 1 byte

   char city[10]="Hyd"; ← string ← 10 bytes

rules:

1. One byte should be left for null char. Otherwise garbage values
2. String var size never smaller than string. Otherwise error
3. We can’t copy string with = operator. We have to use strcpy
4. We can’t compare two strings with ==. We have to use strcmp

Void = empty/nothing

Primitive data types

derived

User defined